

Final-Covid-19

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Import Packages

```
# Add libraries
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr 1.1.4 v readr 2.1.4
## v forcats 1.0.0 v stringr 1.5.1
## v ggplot2 3.4.4 v tibble 3.2.1
## v lubridate 1.9.3 v tidyr 1.3.0
## v purrr 1.0.2

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()

## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(readr)
library(dplyr)
library(lubridate)
library(ggplot2)
```

Import the Data

Copy the link address of the csv file from github.

```
## [1]
"https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_confirmed_global.csv"

## [2]
"https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_deaths_global.csv"

## [3]
```

```
"https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_deaths_US.csv"
```

```
## [4]
```

```
"https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_confirmed_US.csv"
```

Assigning and reading in the data to Variables.

```
# `read_csv()` used to read in the data to variables
```

```
global_cases = read_csv(urls[1])
```

```
global_deaths = read_csv(urls[2])
```

```
# first rows of csv files get better understanding for tidy
```

```
head(global_cases)
```

```
## # A tibble: 6 x 1,147
```

```
## 'Province/State' 'Country/Region' Lat Long '1/22/20' '1/23/20' '1/24/20'
```

```
## <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
```

```
## 1 <NA> Afghanistan 33.9 67.7 0 0 0
```

```
## 2 <NA> Albania 41.2 20.2 0 0 0
```

```
## 3 <NA> Algeria 28.0 1.66 0 0 0
```

```
## 4 <NA> Andorra 42.5 1.52 0 0 0
```

```
## 5 <NA> Angola -11.2 17.9 0 0 0
```

```
## 6 <NA> Antarctica -71.9 23.3 0 0 0
```

```
head(global_deaths)
```

```
## # A tibble: 6 x 1,147
```

```
## 'Province/State' 'Country/Region' Lat Long '1/22/20' '1/23/20' '1/24/20'
```

```
## <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
```

```
## 1 <NA> Afghanistan 33.9 67.7 0 0 0
```

```
## 2 <NA> Albania 41.2 20.2 0 0 0
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```

```
## 5 <NA> Angola -11.2 17.9 0 0 0
```

```
## 6 <NA> Antarctica -71.9 23.3 0 0 0
```

Tidy Data

Tidy the Data — Data reshaping, Conversion, and Renaming

- Pivot the wide-format data (`global_deaths` and `global_cases`) for dates, converting it to a long format.
- Sum totals for each state (`Province/State`) and country (`Country/Region`) for each date.
- Create two new data frames: `covid_global_deaths` for cumulative deaths and `covid_global_cases` for cumulative cases.
- Convert the date columns in both data frames to datetime objects using the `mdy` function from the `lubridate` package.
- Rename columns from `Province/State` to `State` and from `Country/Region` to `Country` in both data frames.

```
# Pivot wide-format data for dates and sum totals for each state
covid_global_deaths = global_deaths %>%
pivot_longer(cols = 13:ncol(global_deaths), names_to = "date") %>%
group_by(`Country/Region`, `Province/State`, date) %>%
summarise("cumulative_deaths" = sum(value, na.rm = TRUE), .groups = 'drop')

covid_global_cases = global_cases %>%
pivot_longer(cols = 13:ncol(global_cases), names_to = "date") %>%
group_by(`Country/Region`, `Province/State`, date) %>%
summarise("cumulative_cases" = sum(value, na.rm = TRUE), .groups = 'drop')

# Convert dates to datetime object
covid_global_deaths$date = lubridate::mdy(covid_global_deaths$date)
covid_global_cases$date = lubridate::mdy(covid_global_cases$date)

# Rename columns from Province_State -> State & Admin2 -> County
covid_global_deaths = covid_global_deaths %>%
rename_at('Province/State', ~'State') %>%
rename_at('Country/Region', ~'Country')

covid_global_cases = covid_global_cases %>%
rename_at('Province/State', ~'State') %>%
rename_at('Country/Region', ~'Country')
```

Merging the Data

Merge the `covid_global_deaths` and `covid_global_cases` data frames based on common columns using the `merge` function. The `all.x=TRUE` argument indicates a left join, meaning all rows from the left data frame (`covid_global_deaths`) will be included.

Create subsets of the merged data frame for specific countries, namely Switzerland (`ch`) and Germany (`de`).

Create tidy versions of the data frames (**de** and **ch**) by removing the **State** column.

```
# merge global data sets and filter to get data just for Switzerland and for germany
world = merge(x=covid_global_deaths, y=covid_global_cases, all.x=TRUE)

ch <- world[world$Country == "Switzerland", ]
de <- world[world$Country == "Germany", ]

de_tidy <- de %>% select(-State)
ch_tidy <- ch %>% select(-State)

# View first several lines of each data set
head(ch_tidy)
head(de_tidy)
```

Add Visualizations and Analysis

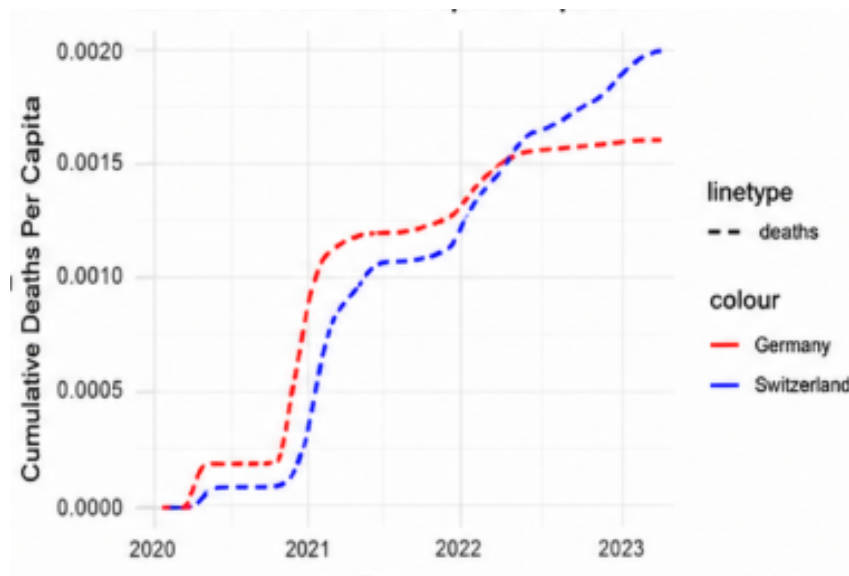
More than half of the population of Switzerland (8.703 million) had covid. while less than half of Germany's population (83.2 million) had covid. And .203% of the population died from covid in germany, while Switzerland had .163% of population. This is interesting, you'd think that the more cases per capita would result in more deaths per capita.

```
# Adding a per capita column to both data sets and viewing their summaries
de_per_cap <- de_tidy %>%
mutate(
  de_deaths_per_capita = cumulative_deaths / 83.2e6,
  de_cases_per_capita = cumulative_cases / 83.2e6
)

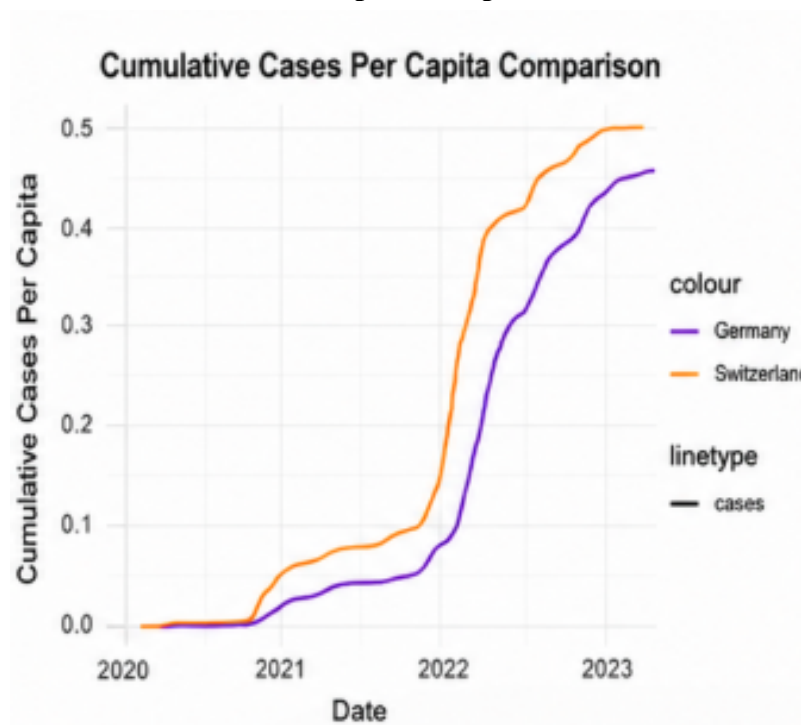
ch_per_cap <- ch_tidy %>%
mutate(
  ch_deaths_per_capita = cumulative_deaths / 8703000,
  ch_cases_per_capita = cumulative_cases / 8703000
)

summary(de_per_cap)
summary(ch_per_cap)
```

Cumulative Deaths Per Capita Comparison



Cumulative Cases Per Capita Comparison



Below we can see Germany did much better than Switzerland in respect to the numbers of deaths per case, but as the pandemic continued Switzerland dropped their deaths per case below Germany.

```
# Calculate deaths per case with a check for division by zero
```

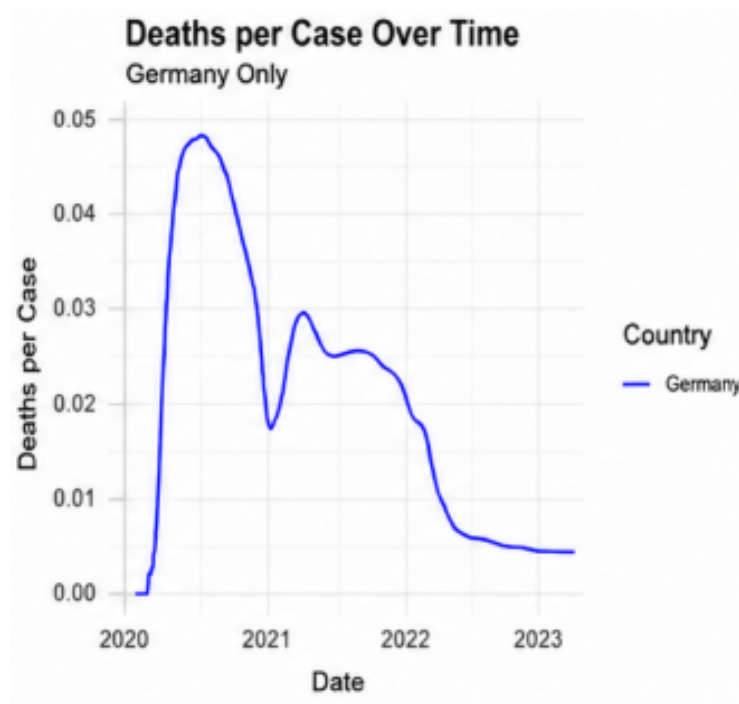
```

de_tidy$deaths_per_case <- ifelse(de_tidy$cumulative_cases > 0,
de_tidy$cumulative_deaths / de_tidy$cumulative_cases,
0)

# Create a time series plot
ggplot(de_tidy, aes(x = date, y = deaths_per_case, color = Country)) +
  geom_line() +
  labs(title = "Deaths per Case Over Time",
x = "Date",
y = "Deaths per Case",
color = "Country") +
  theme_minimal()

```

Germany Deaths per Case Over Time



```

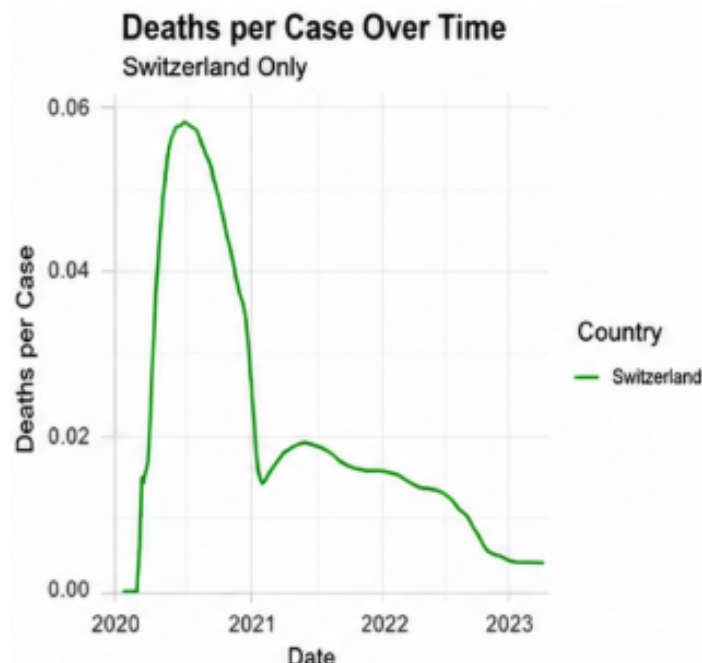
ch_tidy$deaths_per_case <- ifelse(ch_tidy$cumulative_cases > 0,
ch_tidy$cumulative_deaths / ch_tidy$cumulative_cases,
0)

# Create a time series plot
ggplot(ch_tidy, aes(x = date, y = deaths_per_case, color = Country)) +
  geom_line() +
  labs(title = "Deaths per Case Over Time",
x = "Date",
y = "Deaths per Case",

```

```
color = "Country") +  
theme_minimal()
```

Switzerland Deaths per Case Over Time



```
combined_df <- rbind(transform(de_tidy, dataset = "de_tidy"),  
transform(ch_tidy, dataset = "ch_tidy"))  
  
# Create a time series plot with multiple datasets  
ggplot(combined_df, aes(x = date, y = deaths_per_case, color = Country, linetype = dataset)) +  
geom_line() +  
labs(title = "Deaths per Case Over Time",  
x = "Date",  
y = "Deaths per Case",  
color = "Country",  
linetype = "Dataset") +  
theme_minimal()
```

Combined Deaths per Case Over Time

